

RESEARCH SCIENTIST

ABOUT ME

I am an engineer with a PhD in the development and application of hyperspectral imaging systems for the long-wave infrared (LWIR). I work across the full stack of these systems, from optics and experimental work through physical modeling and inverse problems to the software that processes the data.

I also implement AI and machine learning solutions and build data processing pipelines that turn large datasets into usable results. I have a keen eye for detail and enjoy presenting science in a clear and easily digestible way.

SKILLS

PROGRAMMING & DATA
Python (NumPy, SciPy, Matplotlib)
Machine learning & optimization
Local LLMs (Ollama)
Data processing pipelines
Scientific data visualization

IMAGING & OPTICS
Hyperspectral & thermal imaging
Optical & system modeling
Inverse problems

DESIGN & 3D
Illustrator, Photoshop, InDesign
Fusion 360

LANGUAGES
Danish (native), English (fluent)

MADS NIBE LARSEN

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WORK EXPERIENCE

RESEARCH SCIENTIST — FEB 2025 - PRESENT
Newtec Engineering A/S | Odense
Developing a hyperspectral long-wave infrared camera across the full stack, from optics and hardware to data analysis, and implementing AI systems.

RESEARCH ENGINEER — JUL 2021 - JAN 2022
Newtec Engineering A/S | Odense
Analyzed hyperspectral image data and developed and tested a hyperspectral long-wave infrared camera.

EDUCATION

INDUSTRIAL PHD IN ENGINEERING — FEB 2022 - FEB 2025
Newtec Engineering A/S & SDU | Odense & Sønderborg
Thesis: Advances in Hyperspectral Long Wave Infrared Imaging: System Modeling and Applications

MSC IN ENGINEERING, PHYSICS AND TECHNOLOGY — SEP 2019 - JUN 2021
University of Southern Denmark | Odense
Weighted average mark: 11.6/12

BSC IN ENGINEERING, PHYSICS AND TECHNOLOGY — SEP 2016 - JUN 2019
University of Southern Denmark | Odense
Weighted average mark: 11.5/12

SELECTED FIRST-AUTHOR PUBLICATIONS

SIGNATURES OF FLAMES BY LONG-WAVE INFRARED HYPERSPECTRAL IMAGING — JAN 2026
Infrared Physics & Technology, Vol. 152, 106211
DOI: 10.1016/j.infrared.2025.106211

LONG-WAVE INFRARED HYPERSPECTRAL IMAGER BASED ON A SCANNING FABRY-PÉROT INTERFEROMETER — APR 2025
Review of Scientific Instruments, Vol. 96, 043705
DOI: 10.1063/5.0242417

CLASSIFICATION OF COMMERCIAL PLASTICS USING HYPERSPECTRAL LONG WAVE INFRARED EMISSION IMAGING — MAR 2025
Spectrochimica Acta Part A, Vol. 329, 125501
DOI: 10.1016/j.saa.2024.125501

CLASSIFICATION OF BLACK PLASTIC TYPES BY HYPERSPECTRAL IMAGING BASED ON LONG-WAVE INFRARED EMISSION SPECTROSCOPY — DEC 2024
Polymer Testing, Vol. 141, 108629
DOI: 10.1016/j.polymertesting.2024.108629

A HYPERSPECTRAL THERMAL IMAGER BASED ON A LOW ORDER SCANNING FABRY-PÉROT INTERFEROMETER — JUN 2024
Proc. SPIE, Vol. 13030, 130300E
DOI: 10.1117/12.3013384